Entropic Regularization and Sinkhorn

Fast differentiable optimal transport through entropy, scaling, and dual potentials.

Gabriel Peyré

CNRS and École normale supérieure, PSL Université

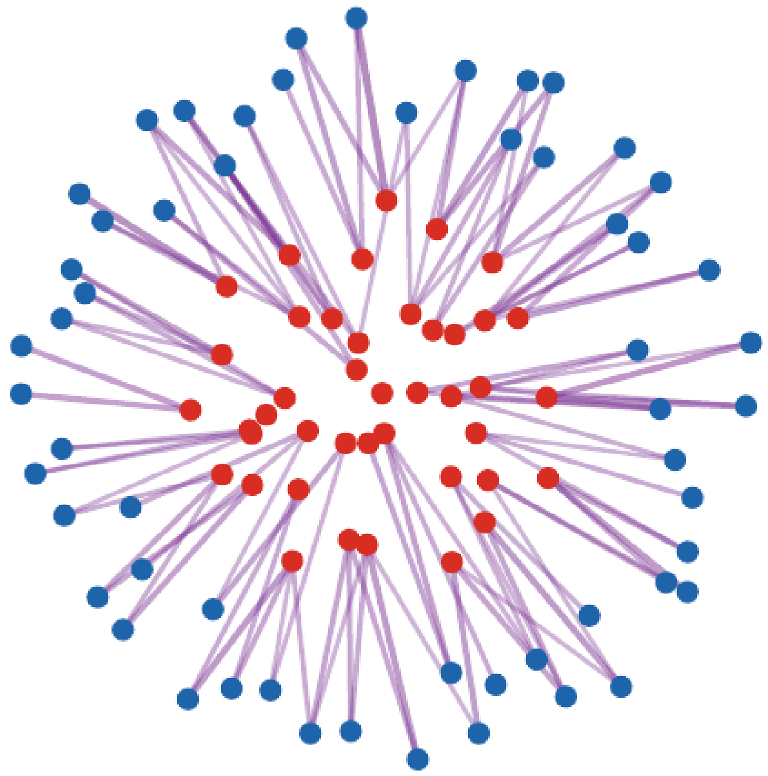
Optimal Transport for Machine Learners

Why regularize OT?

Kantorovich OT is geometrically meaningful but often expensive and non-smooth as a function of the data.

Entropic regularization trades exact sparsity for:

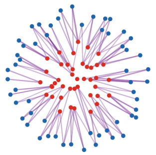
- fast matrix scaling iterations,
- differentiable losses,
- stable dual potentials,
- better statistical behavior at fixed regularization.



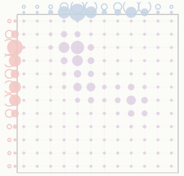
Entropy thickens the transport plan around low-cost pairs.

Roadmap: entropic OT and Sinkhorn

1. Entropic OT

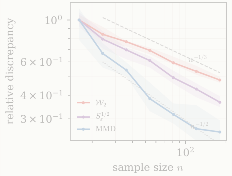


2. Sinkhorn convergence



3. Extensions and debiasing

4. Statistics



5. Learning

Discrete entropic OT

Entropic regularization.
For histograms $a \in \Sigma_n$, $b \in \Sigma_m$, cost matrix C, and $\varepsilon > 0$,

$$\mathcal{L}_{C,\varepsilon}(a,b) = \min_{P \in U(a,b)} \langle C, P \rangle + \varepsilon \sum_{i,j} P_{i,j} \big(\log P_{i,j} - 1 \big),$$

where $U(a,b) = \{P \geq 0 : P\mathbb{1}_m = a,\ P^\top \mathbb{1}_n = b\}$.

The entropy term makes the objective strictly convex on the positive orthant.

Relative entropy form

KL form.
Using the product reference $a \otimes b$, the same minimization can be written, up to an additive constant independent of P, as

$$\min_{P \in U(a,b)} \langle C, P \rangle + \varepsilon \, \mathrm{KL}(P \mid a \otimes b),$$

where

$$\mathrm{KL}(P \mid Q) = \sum_{i,j} P_{i,j} \log \frac{P_{i,j}}{Q_{i,j}} - P_{i,j} + Q_{i,j}.$$

This is the discrete static Schroedinger problem.

Probabilistic interpretation

Let (X, Y) have joint law P and marginals a, b. Then

$$\mathrm{KL}(P \mid a \otimes b) = I(X; Y),$$

the mutual information between X and Y.

Interpretation.
Entropic OT minimizes expected transport cost while penalizing dependence between the source and target labels.

Small ε allows near-deterministic transport. Large ε pushes toward independence.

Continuous formulation

Continuous entropic OT.

For probability measures α, β,

$$\mathcal{L}_{c,\varepsilon}(\alpha, \beta) = \inf_{\pi \in \Gamma(\alpha,\beta)} \int c(x, y)\, d\pi(x, y) + \varepsilon\, \mathrm{KL}(\pi \mid \alpha \otimes \beta).$$

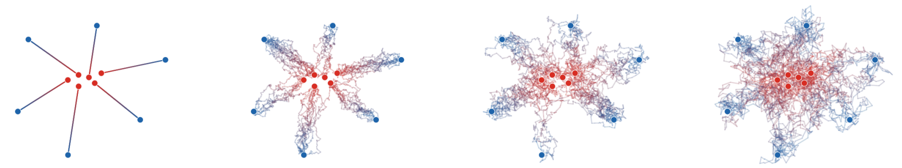
Here $\mathrm{KL}(\pi \mid q) = \int \log(d\pi/dq)\, d\pi$ if $\pi \ll q$, and $+\infty$ otherwise.

Under the usual compactness and lower-semicontinuity assumptions, the value converges to the Kantorovich cost as $\varepsilon \to 0$.

Schroedinger path-space viewpoint

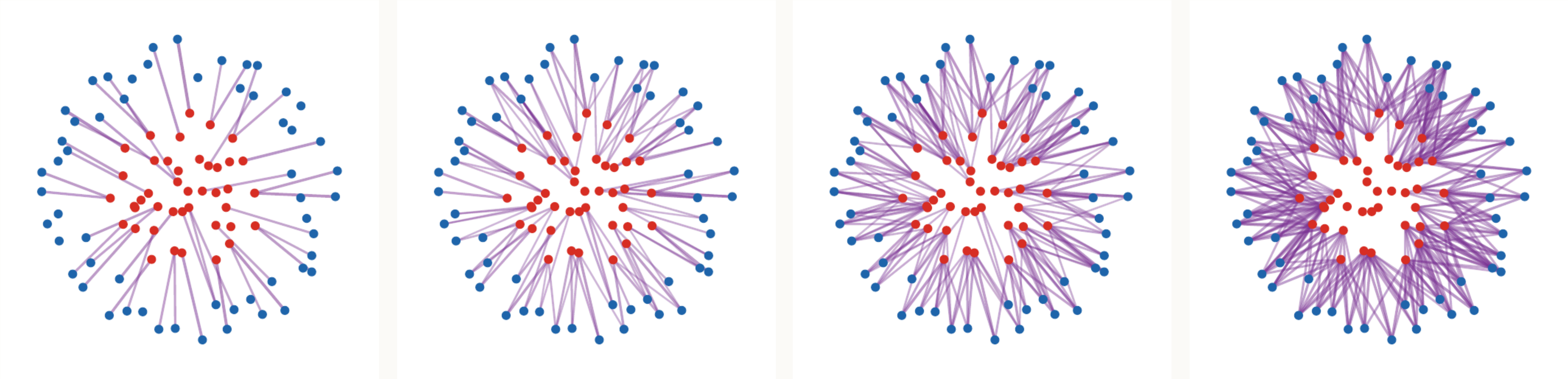
A dynamic version replaces straight-line transport by a controlled Brownian bridge problem. The static coupling is the endpoint law of a path measure.

Static endpoint reduction.
For Brownian reference dynamics, optimizing over path laws with fixed endpoint marginals reduces to an entropic problem over the endpoint coupling.



Increasing ε turns OT rays into Brownian bridges.

Impact of ε



Effect.

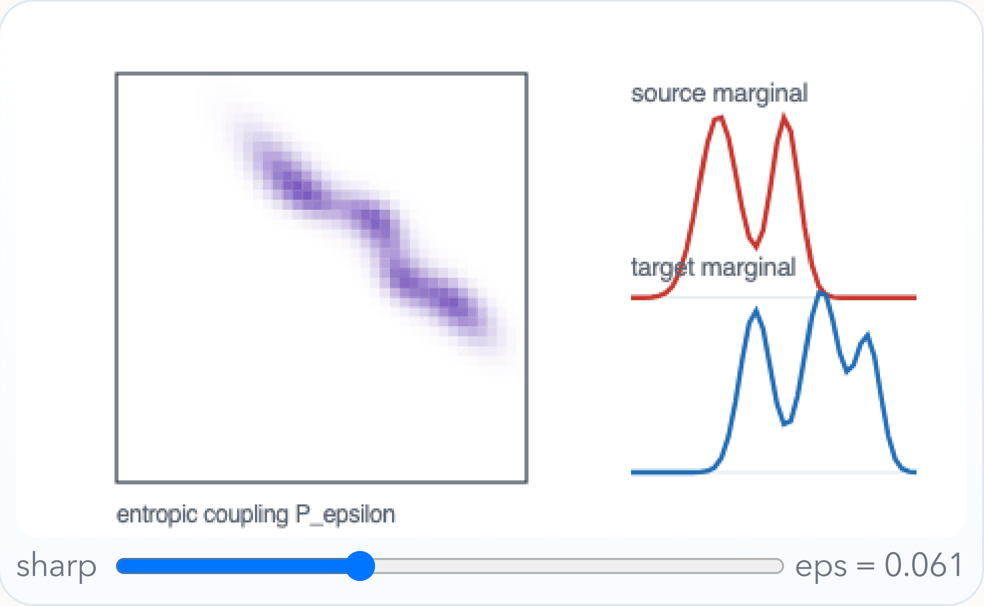
As ε increases, the coupling becomes smoother and less sparse. As ε decreases, it concentrates near the OT support.

Interactive: ε controls the coupling

For $K_\varepsilon = e^{-C/\varepsilon}$, the optimal coupling has the form

$$P_\varepsilon = \operatorname{diag}(u) K_\varepsilon \operatorname{diag}(v).$$

Move the slider to see the coupling sharpen or diffuse.



Sinkhorn scaling form

Scaling form.
Let $K = e^{-C/\varepsilon}$ with positive entries. The entropic optimizer is unique and satisfies

$$P = \mathrm{diag}(u) K \,\mathrm{diag}(v),$$

where $u \in \mathbb{R}^n_+$ and $v \in \mathbb{R}^m_+$ enforce the two marginals. Equivalently,

$$u \odot (Kv) = a, \qquad v \odot (K^\top u) = b.$$

This reduces a constrained convex optimization problem to a matrix scaling problem.

Sinkhorn algorithm

Sinkhorn scaling / IPFP / RAS.

Input: histograms a, b, kernel $K = e^{-C/\varepsilon}$.

Output: scaled coupling P with marginals a, b.

1. Initialize $v^{(0)} = 1_m$.
2. For $\ell = 0, 1, 2, \dots$, update

$$u^{(\ell+1)} = \frac{a}{Kv^{(\ell)}}, \qquad v^{(\ell+1)} = \frac{b}{K^\top u^{(\ell+1)}}.$$

3. Return $P^{(\ell)} = \mathrm{diag}(u^{(\ell)}) K \,\mathrm{diag}(v^{(\ell)})$.

All divisions are entrywise.

The same matrix-scaling iteration appears under several names: Sinkhorn scaling, IPFP in statistics, and RAS in economics and input-output analysis.

Matrix scaling has a long history

Sinkhorn is one member of a broader matrix-scaling family.

Sinkhorn, IPFP, RAS.

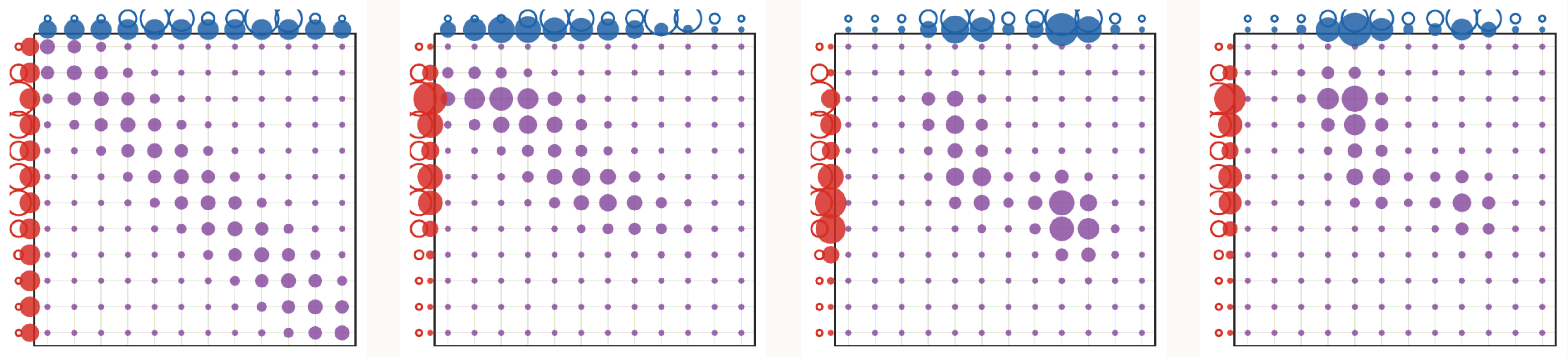
Given a positive matrix K, the goal is to find diagonal scalings $u, v > 0$ such that

$$\mathrm{diag}(u) K \,\mathrm{diag}(v) 1 = a, \qquad \mathrm{diag}(v) K^\top \mathrm{diag}(u) 1 = b.$$

The same alternating normalization appears in statistics, economics, input-output tables, and entropy-regularized OT.

The OT contribution is to choose the matrix as the Gibbs kernel $K = e^{-C/\varepsilon}$.

Sinkhorn half-steps



Each half-step fixes one marginal and breaks the other. Alternating the two projections converges to the intersection.

Computational structure

For dense kernels, one Sinkhorn iteration costs $O(nm)$.

When Sinkhorn is fast.
Sinkhorn is fastest when K has exploitable structure: convolution, separability, low rank, multiscale structure, or GPU-friendly matrix operations.

This explains its role in machine learning: it is differentiable and easy to batch.

Perron-Frobenius viewpoint

Hilbert-metric convergence comes from positive operators, not only from convex analysis.

Positive operator intuition.
A positive linear map contracts the projective geometry of the cone. Sinkhorn alternates two such positive maps, and the resulting contraction explains a linear convergence mechanism when the kernel is strictly positive and well conditioned.

This is the conceptual bridge between classical Perron-Frobenius theory and the Hilbert-metric proof later in this deck.

Other convex regularizers

General regularized OT.
Entropy can be replaced by a convex divergence:

$$\min_{P \in U(a,b)} \langle C, P \rangle + \varepsilon D_\varphi(P \mid a \otimes b).$$

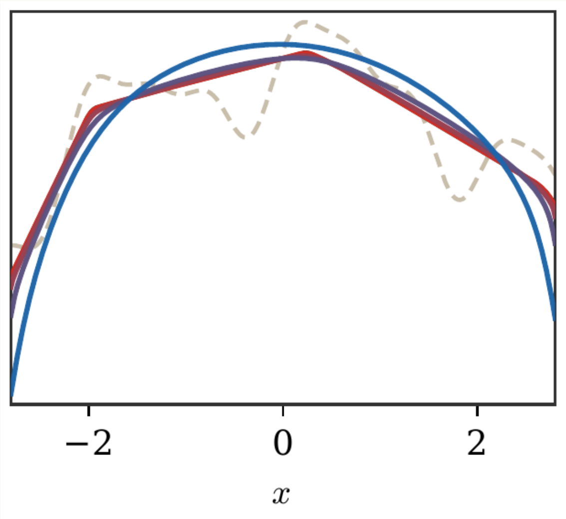
For $Q = a \otimes b$, this means

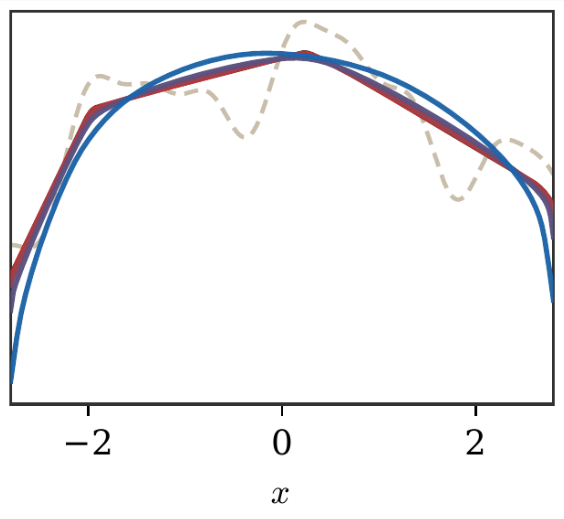
$$D_\varphi(P \mid Q) = \sum_{i,j} Q_{i,j}\, \varphi\left(\frac{P_{i,j}}{Q_{i,j}}\right).$$

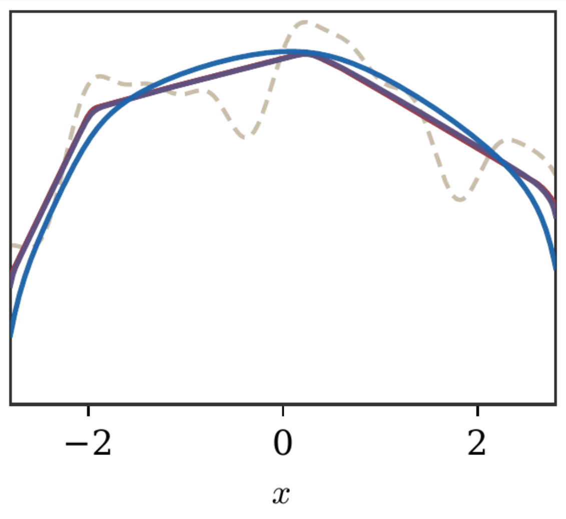
Different choices of φ change both the smoothness and sparsity of the optimal coupling.

Regularizer profiles

Each convex regularizer changes the scalar penalty applied to the density ratio $P_{i,j}/(a_i b_j)$.







Interactive: entropy spreads mass

Entropy replaces a sparse assignment by a Gibbs-weighted cloud around low-cost pairs.

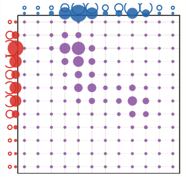
Increasing ε therefore thickens the plan and improves smoothness.

The panel shows the same marginals while only the regularization strength changes.

Roadmap: convergence analysis

1. Entropic OT

2. Sinkhorn convergence



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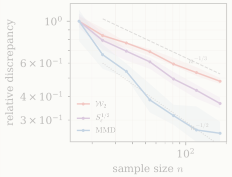


5. Learning

Bregman projection view

Iterative KL projections.
Sinkhorn is alternating Bregman projection for the KL divergence onto the affine marginal constraint sets

$$\mathcal{C}_a = \{P : P\mathbb{1}_m = a\}, \qquad \mathcal{C}_b = \{P : P^\top \mathbb{1}_n = b\}.$$

The initial matrix is the Gibbs kernel $K = e^{-C/\varepsilon}$, and each projection is a multiplicative rescaling.

Dual problem

Entropic dual.
The discrete entropic problem has dual

$$\max_{f,g} \langle f, a\rangle + \langle g, b\rangle - \varepsilon \sum_{i,j} a_i b_j \left(e^{(f_i+g_j-C_{i,j})/\varepsilon} - 1 \right).$$

At optimality,

$$P_{i,j} = a_i b_j e^{(f_i+g_j-C_{i,j})/\varepsilon}.$$

With the convention $P = \mathrm{diag}(u) K \,\mathrm{diag}(v)$, one has $u_i = a_i e^{f_i/\varepsilon}$ and $v_j = b_j e^{g_j/\varepsilon}$.

Soft c-transforms

Soft c-transform.
For a function f on the source support,

$$f^{c,\varepsilon}(y_j) = -\varepsilon \log \sum_i a_i \exp\left(\frac{f_i - C_{i,j}}{\varepsilon}\right).$$

Dual block maximization alternates soft transforms:

$$g \leftarrow f^{c,\varepsilon}, \qquad f \leftarrow g^{\bar{c},\varepsilon}.$$

Stabilized log-domain Sinkhorn

For small ε, $K = e^{-C/\varepsilon}$ can underflow. Stabilized implementations store log-scalings, equivalently dual potentials.

Stabilized log-domain updates.

Instead of scaling vectors u, v, store potentials

$$f_i = \varepsilon \log(u_i/a_i), \qquad g_j = \varepsilon \log(v_j/b_j).$$

The updates are soft c-transforms computed with log-sum-exp operations.

This keeps the Sinkhorn iteration stable when ε is small and the Gibbs kernel has entries below machine precision.

Log-domain update steps

Stabilized Sinkhorn iteration.

Input: histograms a, b, cost matrix C, regularization $\varepsilon > 0$.

Output: stabilized dual potentials f, g.

1. Initialize $f = 0$.
2. Repeat until convergence:

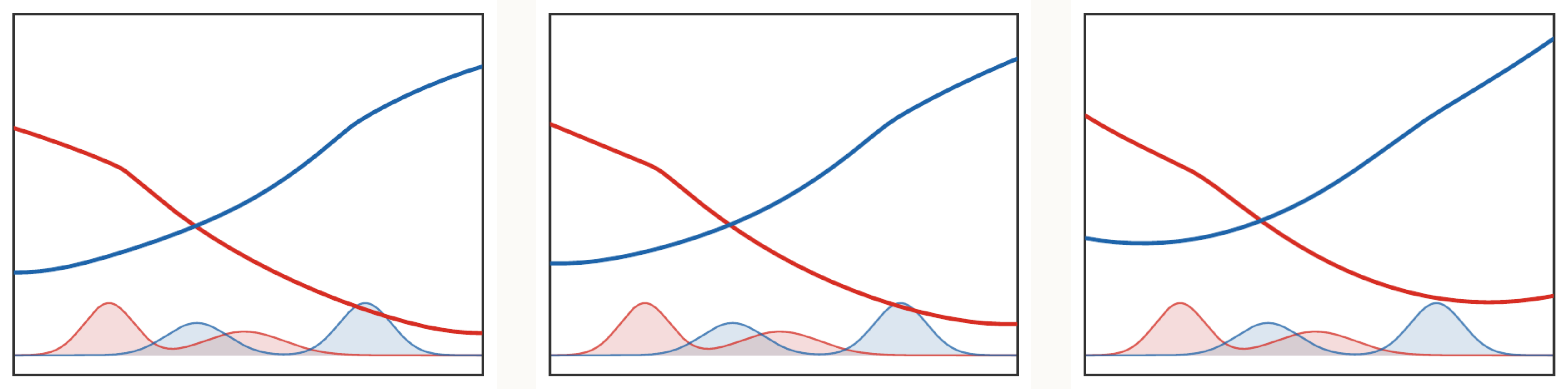
$$g_j \leftarrow -\varepsilon \log \sum_i a_i \exp\left(\frac{f_i - C_{i,j}}{\varepsilon}\right),$$

$$f_i \leftarrow -\varepsilon \log \sum_j b_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right).$$

3. Recover $P_{i,j} = a_i b_j \exp((f_i + g_j - C_{i,j})/\varepsilon)$.

This is numerically stable because it uses log-sum-exp operations.

Dual potentials and ε



The potentials approach unregularized Kantorovich potentials as $\varepsilon \to 0$ and become smoother as ε grows.

Hilbert projective metric

Hilbert metric.

For positive vectors u, v,

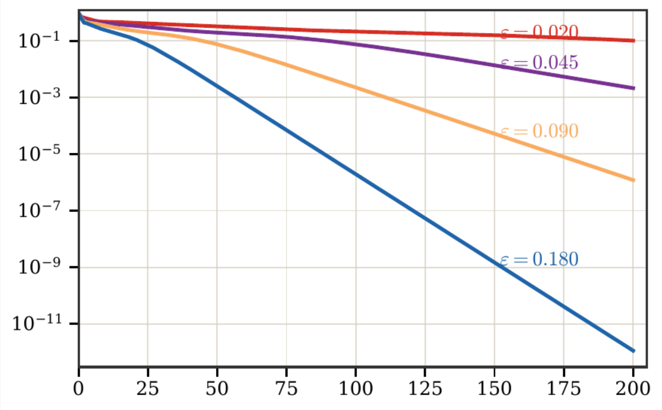
$$d_H(u,v) = \log \max_i \frac{u_i}{v_i} - \log \min_i \frac{u_i}{v_i}.$$

It ignores global scaling. This is exactly what is needed because Sinkhorn scalings are unique only up to reciprocal multiplication.

Linear contraction mechanism

Positive maps contract projectively.
Positive linear maps are nonexpansive in Hilbert metric, and strictly positive matrices contract under quantitative irreducibility assumptions.

Applied to K and $K^\top$, this yields linear convergence of Sinkhorn in a projective metric.



Monotone and variation viewpoints

The log-domain map

$$f \mapsto (f^{c,\varepsilon})^{\bar{c},\varepsilon}$$

is order-preserving and commutes with additive constants.

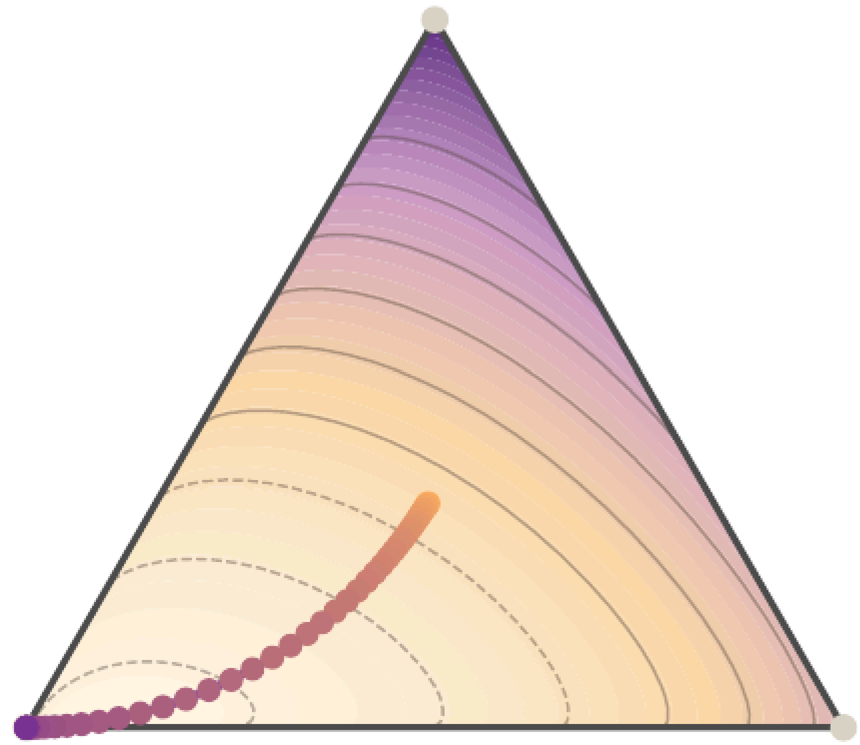
Topical-map intuition.
Order preservation plus translation invariance implies nonexpansiveness for the variation seminorm

$$\|f\|_{\mathrm{var}} = \max_i f_i - \min_i f_i.$$

This gives a robust way to read Sinkhorn as a monotone fixed-point iteration.

Geometry of the regularized path

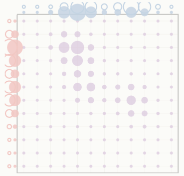
Entropy acts like a barrier inside the transport polytope. The solution follows a central path as ε decreases.



Roadmap: extensions and debiasing

1. Entropic OT

2. Sinkhorn convergence



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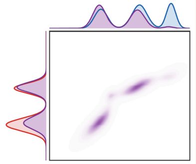
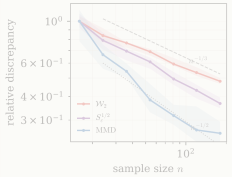


4. Statistics



5. Learning

Unbalanced entropic OT

KL-unbalanced formulation.
Instead of enforcing exact marginals, penalize deviations:

$$\min_{P\geq 0}\langle C,P\rangle+\varepsilon\,\mathrm{KL}(P\mid a\otimes b)+\tau\,\mathrm{KL}(P\mathbb{1}\mid a)+\tau\,\mathrm{KL}(P^\top\mathbb{1}\mid b).$$

This allows mass creation and destruction when matching every unit of mass is unnatural.

Generalized Sinkhorn for unbalanced OT

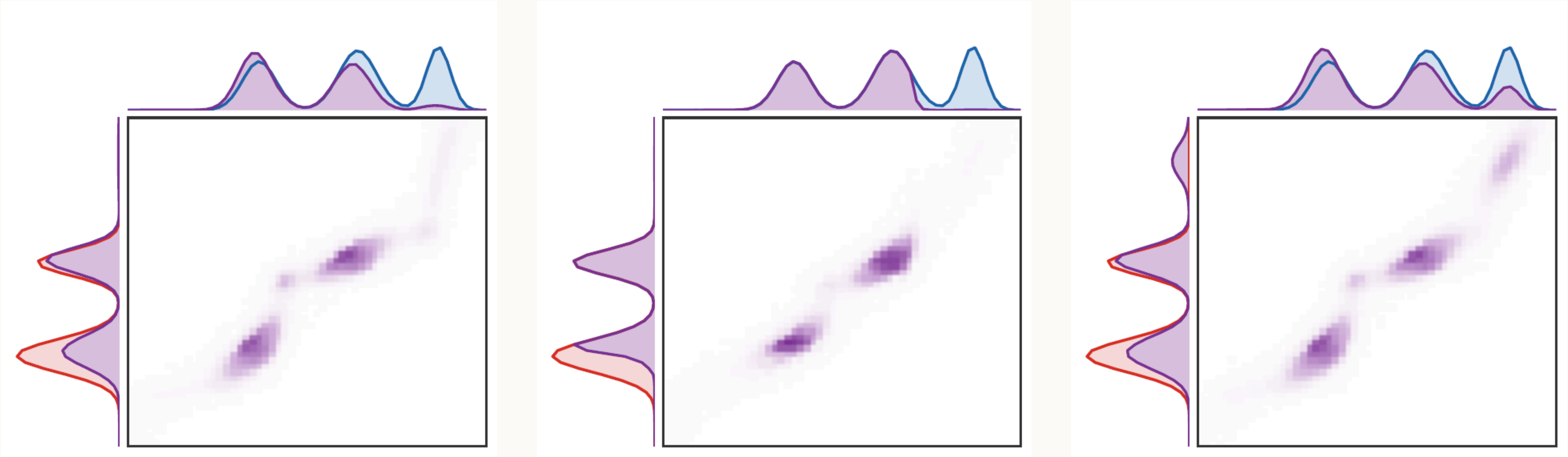
For the KL marginal penalty, the updates become damped:

$$u \leftarrow \left(\frac{a}{Kv}\right)^{\tau/(\tau+\varepsilon)}, \qquad v \leftarrow \left(\frac{b}{K^\top u}\right)^{\tau/(\tau+\varepsilon)}.$$

Balanced limit.

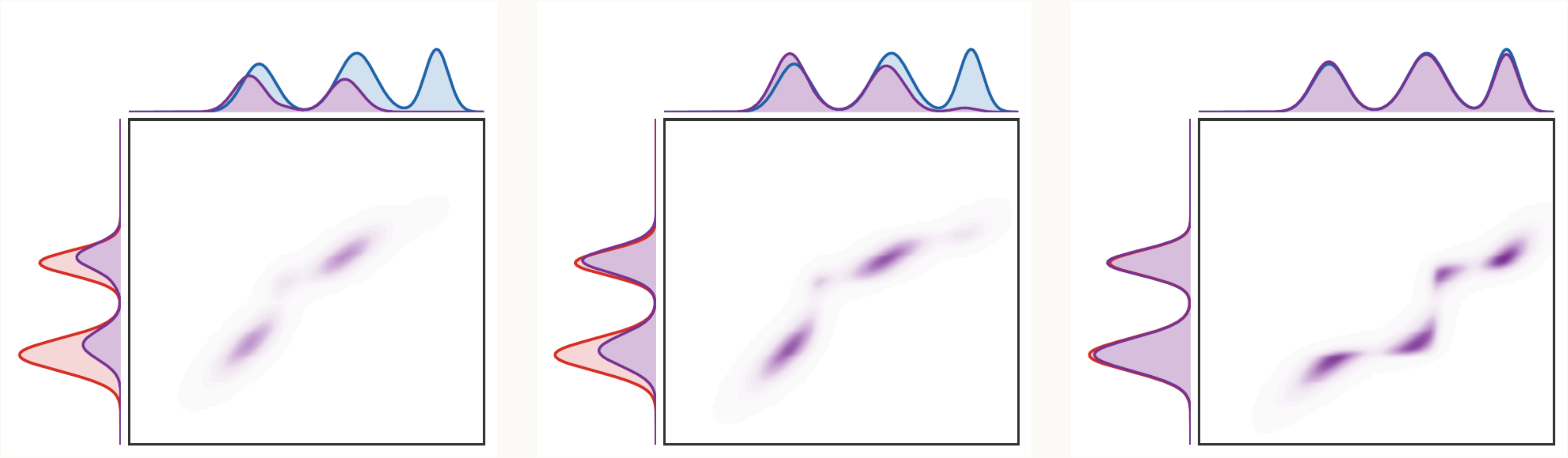
As $\tau \to +\infty$, the exponent tends to one and one recovers ordinary Sinkhorn scaling.

Marginal penalties matter



Different divergences encode different ways of ignoring or creating mass.

Balanced versus unbalanced interpolation



Small marginal penalty favors local mass variation; large penalty approaches balanced OT.

Entropic barycenters

Entropic Wasserstein barycenter.
Given input measures $(\beta_s)_s$ and weights $(\lambda_s)_s$, compute

$$\min_{\alpha} \sum_s \lambda_s \mathcal{L}_{c,\varepsilon}(\alpha, \beta_s).$$

Sinkhorn-like updates alternate between transport couplings and the barycenter weights.

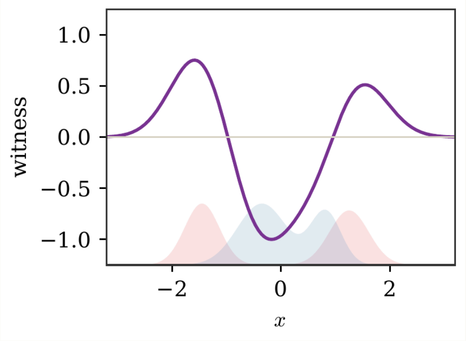
Kernel norms and MMD

MMD.

For a reproducing kernel Hilbert space $\mathcal{H}$,

$$\mathrm{MMD}_k(\alpha, \beta) = \sup_{\|f\|_{\mathcal{H}} \leq 1} \int f \, d(\alpha - \beta).$$

MMD is not an OT distance, but it is another differentiable discrepancy with favorable sample complexity.



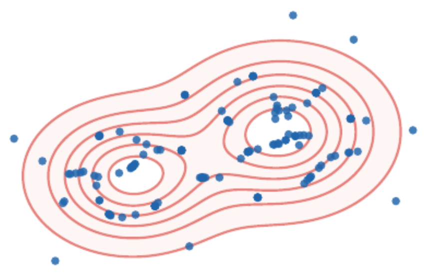
Sinkhorn divergence

Debiased Sinkhorn loss.
The Sinkhorn divergence is

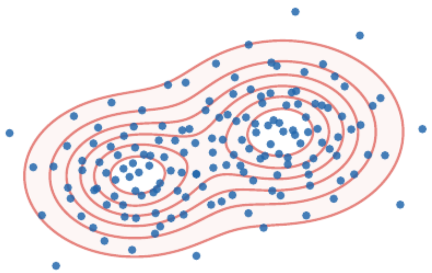
$$S_\varepsilon(\alpha, \beta) = \mathcal{L}_{c,\varepsilon}(\alpha, \beta) - \frac{1}{2}\mathcal{L}_{c,\varepsilon}(\alpha, \alpha) - \frac{1}{2}\mathcal{L}_{c,\varepsilon}(\beta, \beta).$$

The correction removes the entropic self-bias: $S_\varepsilon(\alpha, \alpha) = 0$, while preserving differentiability.

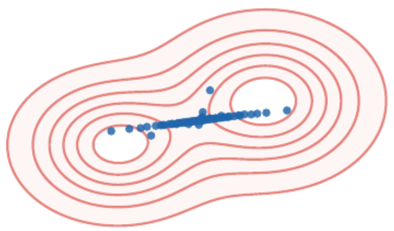
Debiasing effect



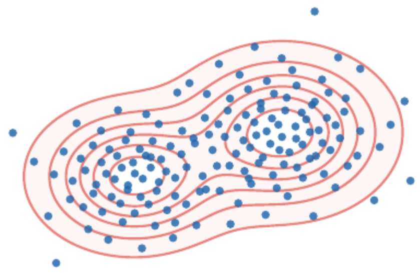
Raw, small ε.



Debiased, small ε.



Raw, large ε.



Debiased, large ε.

Positivity and limits

Sinkhorn divergence behavior.

For many standard costs, S_ε is nonnegative and separates probability measures. Moreover,

$$S_\varepsilon(\alpha, \beta) \to W_c(\alpha, \beta) \quad \text{as } \varepsilon \to 0,$$

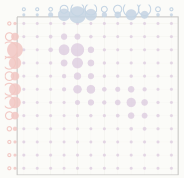
while large ε connects to kernel norms and MMD-type limits.

This gives a continuum between OT geometry and kernel discrepancies.

Roadmap: statistical behavior

1. Entropic OT

2. Sinkhorn convergence



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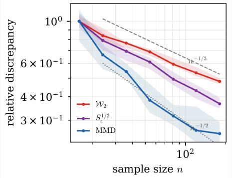


5. Learning

Curse of dimensionality for exact OT

For empirical measures $\hat{\alpha}_n$, exact OT suffers from dimension-dependent rates:

$$\mathbb{E}\, W_p(\hat{\alpha}_n, \alpha) \asymp n^{-1/d}$$

in the high-dimensional absolutely continuous regime, up to known endpoint variations.

Message.
Exact OT is geometrically rich but statistically expensive in high dimension.

Fixed ε changes the statistics

At fixed $\varepsilon > 0$, entropic OT and Sinkhorn divergences are smoother functionals of the empirical measure.

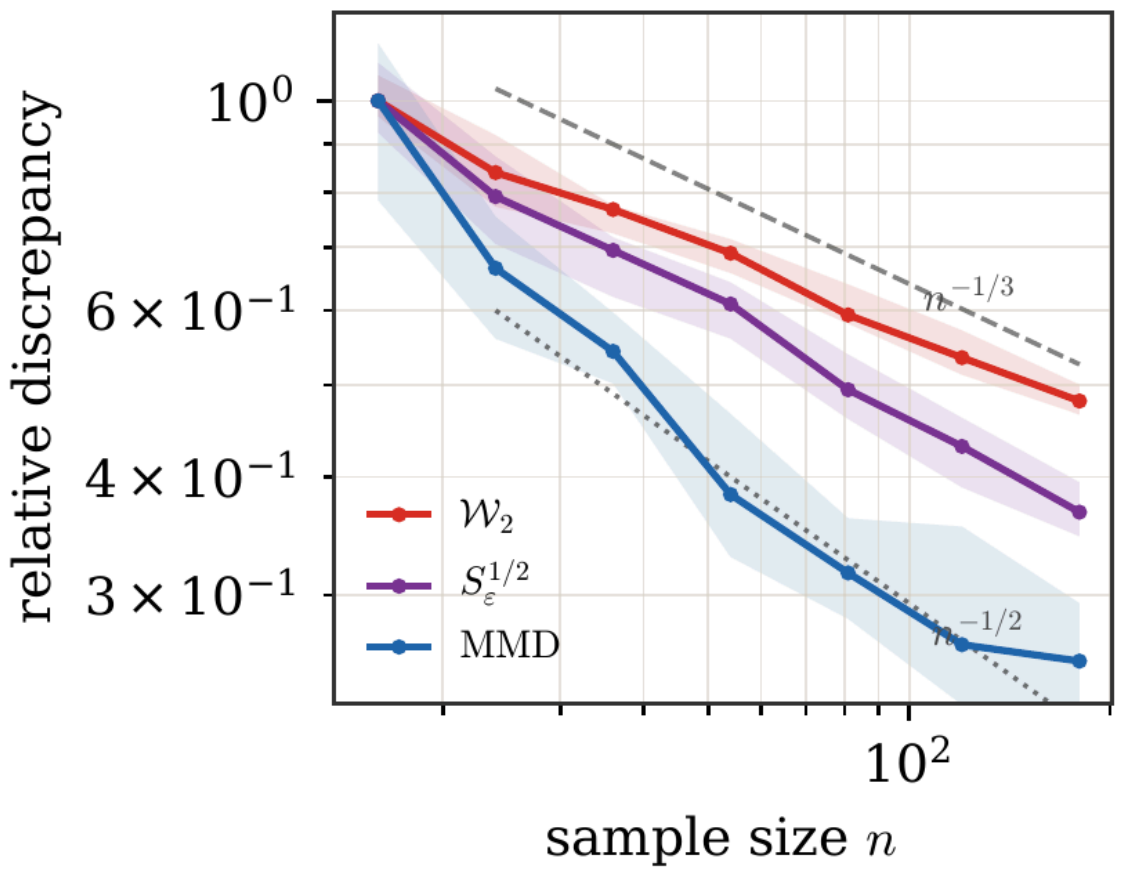
Parametric behavior at fixed ε.

Under regularity assumptions, empirical Sinkhorn divergences admit $n^{-1/2}$-type fluctuations, with constants depending on ε.

The price is bias: fixed ε does not equal exact OT.

Bias-variance tradeoff

Small ε reduces bias but makes computation and statistics harder. Large ε smooths the problem but changes the geometry.



Entropy estimates exact OT through three errors

For empirical data, the approximation to exact OT contains:

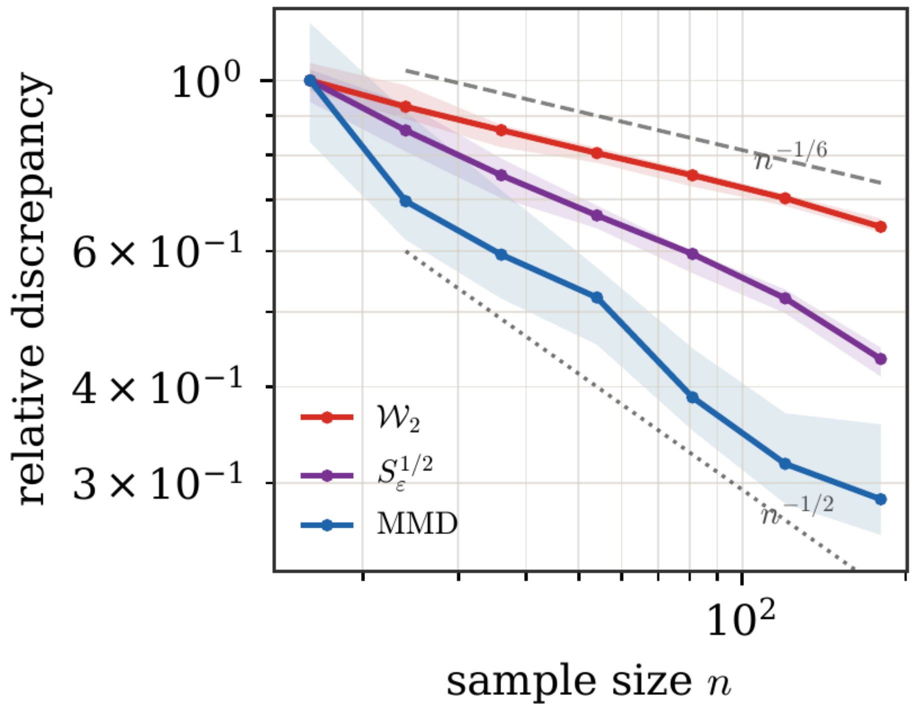
1. sampling error,
2. regularization bias,
3. numerical optimization error.

Practical rule.
Choose ε large enough for stable estimation and fast Sinkhorn iterations, but small enough to retain the desired OT geometry.

Smoothness and high dimension

If the inputs have smooth densities, smoothing or regularization can exploit this smoothness. In high dimension, generic smoothing is itself costly.

This is one reason entropic OT is attractive in practice: it provides a differentiable approximation with scalable linear-algebra structure.



Roadmap: learning with Sinkhorn

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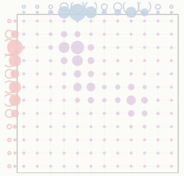
2. Sinkhorn convergence



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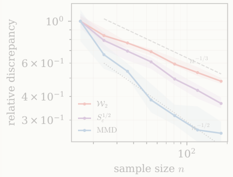
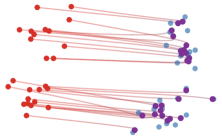


5. Learning



Generative model fitting

Sinkhorn training loss.
For a generator g_θ and latent law ζ, train with

$$\min_\theta S_\varepsilon\big((g_\theta)_\#\zeta, \beta\big).$$

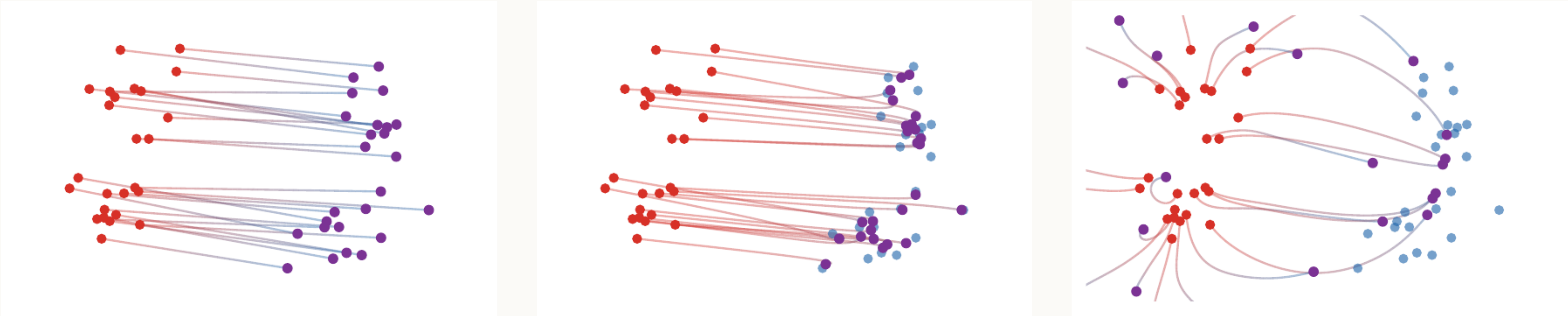
The differentiability of the entropic problem makes this compatible with automatic differentiation.

Discriminator versus generator viewpoint

Kantorovich dual potentials play the role of smooth discriminators:

$$\nabla_\theta S_\varepsilon((g_\theta)_\# \zeta, \beta)$$

can be computed by differentiating through the generator and using the optimal dual potentials or coupling.



Automatic differentiation

Envelope principle.
Once the optimal dual variables are computed, the derivative of the regularized transport value with respect to the cost is the optimal coupling:

$$\frac{\partial \mathcal{L}_{C,\varepsilon}}{\partial C_{i,j}} = P^{\star}_{i,j}.$$

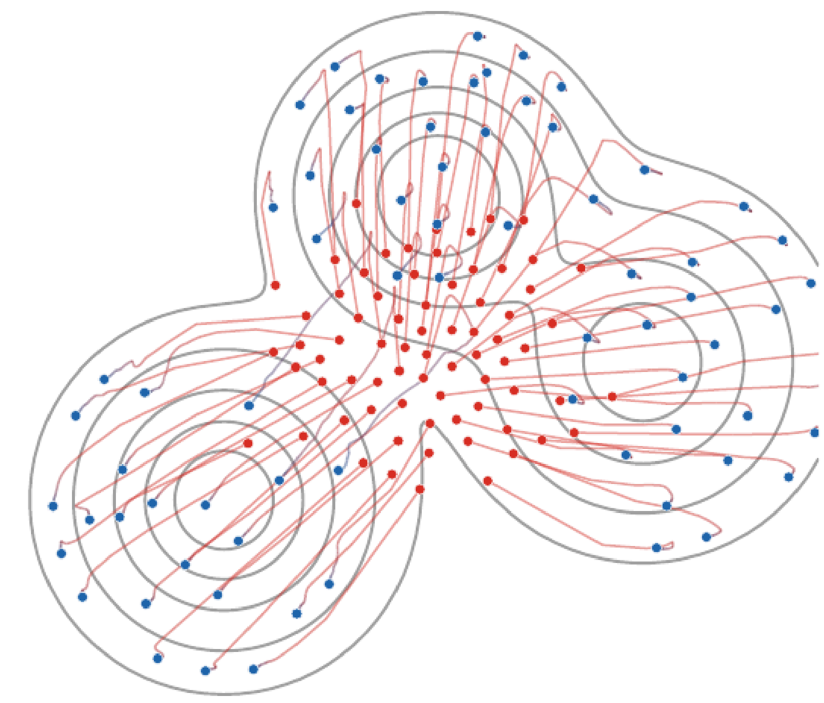
With respect to weights, the derivatives are the optimal dual potentials, up to the usual additive normalization.

This is the basic reason Sinkhorn losses integrate well into differentiable learning pipelines.

Training architecture

A typical mini-batch pipeline is:

1. sample latent variables and data,
2. build the cost matrix,
3. run Sinkhorn iterations,
4. backpropagate the Sinkhorn loss through the generator.



Open problems and design tensions

Main tension.
Entropic OT is an approximation, a numerical method, a statistical smoother, and a differentiable loss. The best ε depends on the role.

Open directions include scalable approximations in high dimension, adaptive regularization, metric learning for costs, and robust generative training.

Summary

Entropy
smooths OT plans

Sinkhorn
solves by scaling

Debiasing
turns costs into losses

Takeaway.
Entropic regularization transforms optimal transport into a scalable differentiable primitive for optimization, statistics, and learning.

Next deck

The next presentation focuses on duality, semi-discrete OT, c-transforms, and Wasserstein-1 type dual norms.